Classical Field Theory — Exercise Sheet 10

Residual gauges and charged tails

Hints and answers

Write $\Box = \partial_t^2 - \Delta$. Use $\mathbf{E} = -\nabla\phi_{\mathrm{em}} - \partial_t \mathbf{A}$, $\mathbf{B} = \operatorname{curl}\mathbf{A}$ and $\ell = -(\partial_t\phi_{\mathrm{em}} + \operatorname{div}\mathbf{A})$. Lecture 10's theorem assumes a smooth conserved source of compact *spacetime* support and compatible compact smooth potential data. Problems 2–3 explicitly separate a charged background outside that class.

Exercise 10.1 (All residual shifts of Cauchy data). Let a smooth Lorenz-gauge Maxwell potential have initial data $(a_\phi, b_\phi, a_A, b_A)$ as in Lecture 10, with a fixed source ρ. Let χ be the smooth wave with arbitrary real compact smooth data $\chi(0) = p$, $\chi_t(0) = q$. Set $\phi'_{\mathrm{em}} = \phi_{\mathrm{em}} - \chi_t$, $\mathbf{A}' = \mathbf{A} + \nabla\chi$.

(a) Find the increments of all four potential data. Verify directly that both compatibility expressions $b_\phi + \operatorname{div} a_A$ and $-\Delta a_\phi - \operatorname{div} b_A - \rho(0)$ are unchanged. Compare the two initial electromagnetic fields.

(b) Prove that fixing all four potential data rules out every nonzero residual gauge wave in this compact-data class. Contrast this with fixing only the electromagnetic fields. If compactness of the gauge data is dropped, explain why a constant χ is physically trivial.

Hint. Use $\chi_{tt}(0) = \Delta p$ before evaluating the scalar-potential velocity. The vector displacement and velocity involve different gauge data.

Answer. The increments are $(-q, -\Delta p, \nabla p, \nabla q)$. Both compatibility expressions and both field data are unchanged. Unchanged potential data force $p = q = 0$ in the compact-data class, hence $\chi = 0$ by wave uniqueness; a global constant has $\mathrm{d}\chi = 0$.

Exercise 10.2 (A smooth charge with a Coulomb tail). Choose a radial nonnegative $b \in C_c^\infty(\mathbb{R}^3)$ with $\int b\,\mathrm{d}^3x = 1$ and support in $\overline{B_R(0)}$, $R > 0$. Fix real $Q \neq 0$ and prescribe the static source $\rho_Q(x) = Qb(x)$, $\mathbf{J} = 0$ for all time. For $r > 0$ define, writing $\rho_Q(s)$ for its radial profile,

$$M_Q(r) = 4\pi \int_0^r s^2 \rho_Q(s)\mathrm{d}s, \quad E_r(r) = \frac{M_Q(r)}{4\pi r^2}, \quad \phi_Q(r) = \int_r^\infty E_r(s)\mathrm{d}s.$$

(a) Prove that $E_Q(x) = E_r(|x|)x/|x|$ and $\phi_Q(|x|)$ extend smoothly across zero. Prove Gauss's law there as well as away from zero. Show that $(\phi_{\mathrm{em}}, \mathbf{A}) = (\phi_Q, 0)$ solves Maxwell in Lorenz gauge.

(b) Determine the exact potential and field for $r \geq R$. Prove $\phi_Q \notin L^2$, hence $\phi_Q \notin H^1$, while $\nabla\phi_Q \in L^2$ and the field energy is finite. Calculate the electric energy outside B_R.

(c) Identify the failed hypotheses of Lecture 10's theorem. Why does conservation alone not force $Q = 0$ for this source?

Hint. Remove the radial denominator by proving $E_Q(x) = x\int_0^1 a^2\rho_Q(ax)\mathrm{d}a$. Its divergence is the integral of an a-derivative. For the potential, integrate this smooth field on the segment from zero to x.

Answer. The fields are smooth, with $\phi_Q(r) = Q/(4\pi r)$ and $E_r(r) = Q/(4\pi r^2)$ for $r \geq R$. The potential fails L^2 at infinity, but its gradient has finite norm. Exterior electric energy is $Q^2/(8\pi R)$. The source is spatially compact and conserved, but is not compact in time.

Exercise 10.3 (An affine charged evolution class). Keep the background of Exercise 10.2. Let $(\rho_1, J_1) \in C_c^\infty(\mathbb{R}^{1+3}; \mathbb{R} \times \mathbb{R}^3)$ be conserved. Give compact smooth perturbation data $(a_{1\phi}, b_{1\phi}, a_{1A}, b_{1A})$ satisfying

$$b_{1\phi} + \operatorname{div} a_{1A} = 0, \qquad -\Delta a_{1\phi} - \operatorname{div} b_{1A} = \rho_1(0).$$

Construct a smooth Lorenz-gauge solution for total source $(\rho_Q + \rho_1, J_1)$ and total initial data obtained by adding the static background. Prove its total charge is Q, its electric and magnetic fields have finite energy on each slice, and $\phi_{\mathrm{em}}(t) \in \phi_Q + H^1(\mathbb{R}^3)$ while $\phi_{\mathrm{em}}(t) \notin H^1$. State the uniqueness conclusion actually supplied by Lecture 10. Explain why this construction does not put the charged background in a Schwartz canonical phase space.

Hint. Apply the compact theorem only to the perturbation. Use its retarded construction and finite propagation to keep the perturbation data compact on each bounded time interval. Add the two Maxwell solutions afterwards.

Answer. Use Lecture 10 to construct (ϕ_1, A_1) and set $(\phi_{\mathrm{em}}, \mathbf{A}) = (\phi_Q + \phi_1, A_1)$. The perturbation, hence the total field after subtracting the fixed background, is unique in the theorem's C^2 Lorenz class with the specified potential data. The scalar potential belongs to the affine class $\phi_Q + H^1$, not to H^1.

Exercise 10.4 (Local potentials with no global real one-form). On the oriented unit sphere, with area form $\sin\theta \mathrm{d}\theta \wedge \mathrm{d}\varphi$, let $g_{\mathrm{mag}} \neq 0$ be real. On $U_N = S^2 \setminus \{\text{south pole}\}$ and $U_S = S^2 \setminus \{\text{north pole}\}$, respectively, take

$$A_N = g_{\mathrm{mag}}(1 - \cos\theta)\mathrm{d}\varphi, \qquad A_S = -g_{\mathrm{mag}}(1 + \cos\theta)\mathrm{d}\varphi.$$

These are spatial real one-form potentials; the task uses exterior calculus from *Topology and Geometry*, Part I, not bundle theory as a prerequisite.

(a) Verify smoothness at each included pole, compute the common curvature $F = \mathrm{d}A_N = \mathrm{d}A_S$, and find its total flux.

(b) Find a gauge function on a simply connected azimuthal sector in the overlap. Prove that there is no single-valued real gauge function for this shift on the whole overlap, and no globally smooth real one-form A on S^2 with $\mathrm{d}A = F$.

The interpretation by nontrivial bundles and single-valued charged transition phases, including the allowed flux quantization, is a later topic: *Topology and Geometry*, Lectures 20–22. No bundle construction or quantization theorem is assumed in this exercise.

Hint. Put $X = \sin\theta\cos\varphi$, $Y = \sin\theta\sin\varphi$ and $Z = \cos\theta$. Rewrite $\mathrm{d}\varphi$ using $X\mathrm{d}Y - Y\mathrm{d}X$. Test exactness on the equatorial circle and then apply Stokes on the closed sphere.

Answer. $F = g_{\mathrm{mag}} \sin\theta \mathrm{d}\theta \wedge \mathrm{d}\varphi$ has flux $4\pi g_{\mathrm{mag}} \neq 0$. Locally $A_N - A_S = \mathrm{d}(2g_{\mathrm{mag}}\varphi)$, but its equatorial integral is $4\pi g_{\mathrm{mag}}$, excluding a single-valued real gauge function on the overlap and a global real potential on the sphere.